

# Installing Beliefs to Audit Attribution: A Ground-Truth Testbed and Five Ways It Misleads

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## 1 Measured causal effects as the testbed ground truth

The testbed ground truth is framed as a measured causal effect size rather than a dataset-novelty claim. Its substance is organized around five validation failures, each paired with measured evidence and a detection recipe whose failure labels and recipes map to the atomic results and tables.

## 2 Contrast definitions, scope discipline, and validation rules

Absorption denotes held-out premise specialization rather than a belief measure. Belief and action contrasts use matched off-topic control netting and do not establish causal mediation, while raw belief and action checkpoint figures remain diagnostic rather than netted transfer estimates.

The analysis is exploratory, with model, topic, seed, and checkpoint scope reported only from recorded artifacts. Two-seed scope applies only to declared seed-paired contrast families; single-seed and instrument-specific readings remain separately labeled because the evidence bundle is not uniformly two-seed.

Fixed tables are authoritative for numerical estimates and confidence intervals. The long-sparse cell is interpreted as slow rather than inert, and each belief ratio carries its optimizer-step reading; ratios are omitted where instability or interval structure makes them unsupported.

## 3 Instrument demonstration and five validation failures

The belief instrument is demonstrated by its recorded explicit-stance positive control: it is large at the primary single-seed reading and moderate at the later single-seed reading, where it has begun to decay while the evidence cell is still rising. This is a positive control for the belief instrument rather than general belief transfer; its token dose is lower than the evidence corpus, the point estimates derive from recorded arm scores, and no intervals are inferred.

The descriptive-inference axis is withdrawn as an instrument failure because its recorded explicit-arm positive control is inverted. The byte-identical null-control facet is the highest-moving of all eight facets, so the suite cannot demonstrate premise-specificity and no transfer number from it is reported; the point estimate derives from recorded arm scores, and no interval is inferred.

On the software-architecture topic, the evidence-only belief contrast is near zero and its interval includes zero under a matched off-topic control with all arms gated. This is one of three declared seeds whose netted signs disagree. The seed-seven replicate is also near zero but its interval excludes zero under a seed-matched control. The third-seed replicate is near zero and includes zero with all five arms gated; its sign disagrees with the other declared seeds because of the control machinery term rather than the treatment contrast.

The primary evidence-only belief reading is near zero and its interval excludes zero, despite earlier drafts treating it as the near-null cell; this is a single-seed reading with a matched off-topic control and all arms gated. With the same arms and control, the later single-seed reading is small and excludes zero. All seven arms gated and the control passes, making the gate failure genuinely late.

Across the form cross, the short-dense contrast is moderate and excludes zero across three seeds, but this is only a band-level finding: the premise specification matches the long-form arms and form is the only variable. The matched-density long-dense arm is small and excludes zero. The short-sparse arm is likewise

small and excludes zero, but it is an unstable three-seed corner, so the reading is directional and carries no ratio.

On the model axis, the recorded explicit-stance belief positive control is moderate for the smaller model and small for the larger model, with both intervals excluding zero. The smaller-model arm is dose-matched to the larger-model arm and differs only by model; the larger-model control was retrained, and its separately calibrated gate threshold replaces rather than reuses the smaller-model bar. Action diverges: the smaller-model positive-control contrast is near zero and includes zero, precluding a conduction ratio, whereas the larger-model contrast is small and excludes zero. Its direction replicates at the paired seed, but substantial cross-seed scatter precludes magnitude generalization. Under full fine-tuning, the evidence-corpus belief contrast is near zero and excludes zero in both the first declared seed and its seed-matched paired seed with a re-derived schedule. Because the same corpus under LoRA straddles zero on probability while excluding zero on log-odds, this is interpreted as scale dependence rather than a capacity cliff.

Table 1: Declared contrasts selected or derived from immutable result summaries. Source: paper/synthesis.json.

| intervention                                                           | net effect                 | qualification                                                                                                                                                                                                                                                                                               |
|------------------------------------------------------------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Second topic, evidence-only, belief, seed 42                           | +0.0078 [-0.0096, +0.0252] | Second topic (between architectures), matched off-topic control, all arms gated. Reported as one of three seeds whose initial topic diverges.                                                                                                                                                               |
| Second topic, evidence-only, belief, seed 7                            | +0.0122 [-0.0028, +0.0280] | Second topic (between architectures), matched off-topic control, all arms gated. Reported as one of three seeds whose initial topic diverges.                                                                                                                                                               |
| Second topic, evidence-only, belief, seed 123                          | +0.0104 [-0.0156, +0.0365] | Second topic (between architectures), matched off-topic control, all arms gated. Reported as one of three seeds whose initial topic diverges.                                                                                                                                                               |
| Evidence-only, belief, step 24 (2 epochs)                              | +0.0075 [-0.0026, +0.0176] | The reading the nuclei drifts quoted to the near-null cell. Matched off-topic control, all arms gated.                                                                                                                                                                                                      |
| Evidence-only, belief, step 36 (2 epochs)                              | +0.0342 [-0.0246, +0.0930] | Same arms, same control, three epochs. All seven arms gated; the control passes at 0.012, so the late-step gate failure is genuinely late. One seed.                                                                                                                                                        |
| Explicit stance, belief, step 24                                       | +0.0111                    | Positive control for the belief instrument. Values close to lower than the evidence corpus. Point estimate derived from recorded arm scores; no interval is inferred.                                                                                                                                       |
| Explicit stance, belief, step 36                                       | +0.0247                    | The same positive control one epoch later, where it has begun to drop while the evidence cell is still rising. Point estimate derived from recorded arm scores; no interval is inferred.                                                                                                                    |
| Short, dense (SML), belief                                             | +0.1380 [-0.0001, +0.1497] | Same positive control as the long-form arms, form is the only variable. Dose-matched across three seeds.                                                                                                                                                                                                    |
| Long, dense (SML), belief                                              | +0.0547 [-0.0001, +0.0795] | The length arm of the form cross at matched density.                                                                                                                                                                                                                                                        |
| Short, sparse (SL), belief                                             | +0.0506 [-0.0006, +0.0720] | The variable center, center 2500 across three seeds, so no reading is directional and carries no ratio.                                                                                                                                                                                                     |
| Explicit stance at 40, belief                                          | +0.1517 [-0.1100, +0.1942] | Dose-matched to the 40 arm, model is the only difference.                                                                                                                                                                                                                                                   |
| Explicit stance at 40, action                                          | +0.0070 [-0.0026, +0.0167] | Control retrained at 40. QueryCell has its own calibrated gate threshold; the 40 bar is not reused.                                                                                                                                                                                                         |
| Explicit stance at 40, action                                          | +0.0011 [-0.0018, +0.0040] | Straddle arm. The conduction ratio is quiescent while the response remains.                                                                                                                                                                                                                                 |
| Full fine-tuning, evidence corpus, belief, seed 42                     | +0.0032 [-0.0249, +0.0400] | Each arm. Replicated to direction at seed 7, consistent at percent across seeds, so no magnitude.                                                                                                                                                                                                           |
| Full fine-tuning, evidence corpus, belief, seed 7                      | +0.0141 [-0.0060, +0.0400] | Under full, the same corpus straddle arm on probability and direction are on log-odds, so the contrast is scale-dependent rather than a capacity cliff.                                                                                                                                                     |
| Full fine-tuning, evidence corpus, belief, seed 123                    | +0.0113 [-0.0046, +0.0310] | Seed-seen replication with a seed-matched full-FT control and a re-derived schedule.                                                                                                                                                                                                                        |
| Descriptive stance ratio, explicit arm (the INVERTED positive control) | +0.0020                    | This entry is the last-seen failure mode. On the arm the two-headed self-control level is the highest meeting of all right hands, so the ratio cannot demonstrate greater specificity and no transfer number from it is reported. Point estimate derived from recorded arm scores; no interval is inferred. |

Note. Rows without intervals are deterministic point contrasts over recorded arm scores.

Table 2: Recorded belief contrast and sensitivity quantities. Source: matrix\_v1\_step24/belief\_summary.yaml.

| quantity      | recorded value                    |
|---------------|-----------------------------------|
| delta_raw     | +0.0033 [-0.0009, +0.0082]        |
| machinery     | -0.0039 [-0.0093, +0.0007]        |
| delta_net     | <b>+0.0072</b> [+0.0016, +0.0136] |
| sensitivity   | <b>+0.6521</b> [+0.5578, +0.7413] |
| T (delta_net) | +0.0110                           |

Note. Intervals are paired bootstrap CIs from the stored summary; bold excludes zero.

Table 3: Recorded belief contrast and sensitivity quantities. Source: matrix\_v1\_step36/belief\_summary.yaml.

| quantity      | recorded value                    |
|---------------|-----------------------------------|
| delta_raw     | <b>+0.0228</b> [+0.0154, +0.0307] |
| machinery     | <b>-0.0114</b> [-0.0175, -0.0050] |
| delta_net     | <b>+0.0342</b> [+0.0248, +0.0443] |
| sensitivity   | <b>+0.6521</b> [+0.5578, +0.7413] |
| T (delta_net) | +0.0524                           |

Note. Intervals are paired bootstrap CIs from the stored summary; bold excludes zero.

Table 4: Recorded belief contrast and sensitivity quantities. Source: ms3p\_arms/belief\_summary.yaml.

| quantity      | recorded value                    |
|---------------|-----------------------------------|
| delta_raw     | <b>+0.1034</b> [+0.0741, +0.1352] |
| machinery     | <b>-0.0156</b> [-0.0271, -0.0048] |
| delta_net     | <b>+0.1190</b> [+0.0901, +0.1497] |
| sensitivity   | <b>+0.6521</b> [+0.5578, +0.7413] |
| T (delta_net) | +0.1825                           |

Note. Intervals are paired bootstrap CIs from the stored summary; bold excludes zero.

Table 5: Recorded inference contrast and sensitivity quantities. Source: inference\_v1\_step24/inference\_summary.yaml.

| quantity  | recorded value                    |
|-----------|-----------------------------------|
| delta_raw | <b>+0.0134</b> [+0.0043, +0.0245] |
| machinery | +0.0013 [-0.0049, +0.0075]        |
| delta_net | <b>+0.0121</b> [+0.0012, +0.0245] |

*Note.* Intervals are paired bootstrap CIs from the stored summary; bold excludes zero.

Table 6: Recorded action contrast and sensitivity quantities. Source: h8\_8b/action\_summary.yaml.

| quantity  | recorded value                    |
|-----------|-----------------------------------|
| delta_raw | <b>+0.0201</b> [+0.0128, +0.0280] |
| machinery | <b>-0.0151</b> [-0.0215, -0.0090] |
| delta_net | <b>+0.0352</b> [+0.0249, +0.0459] |

*Note.* Intervals are paired bootstrap CIs from the stored summary; bold excludes zero.

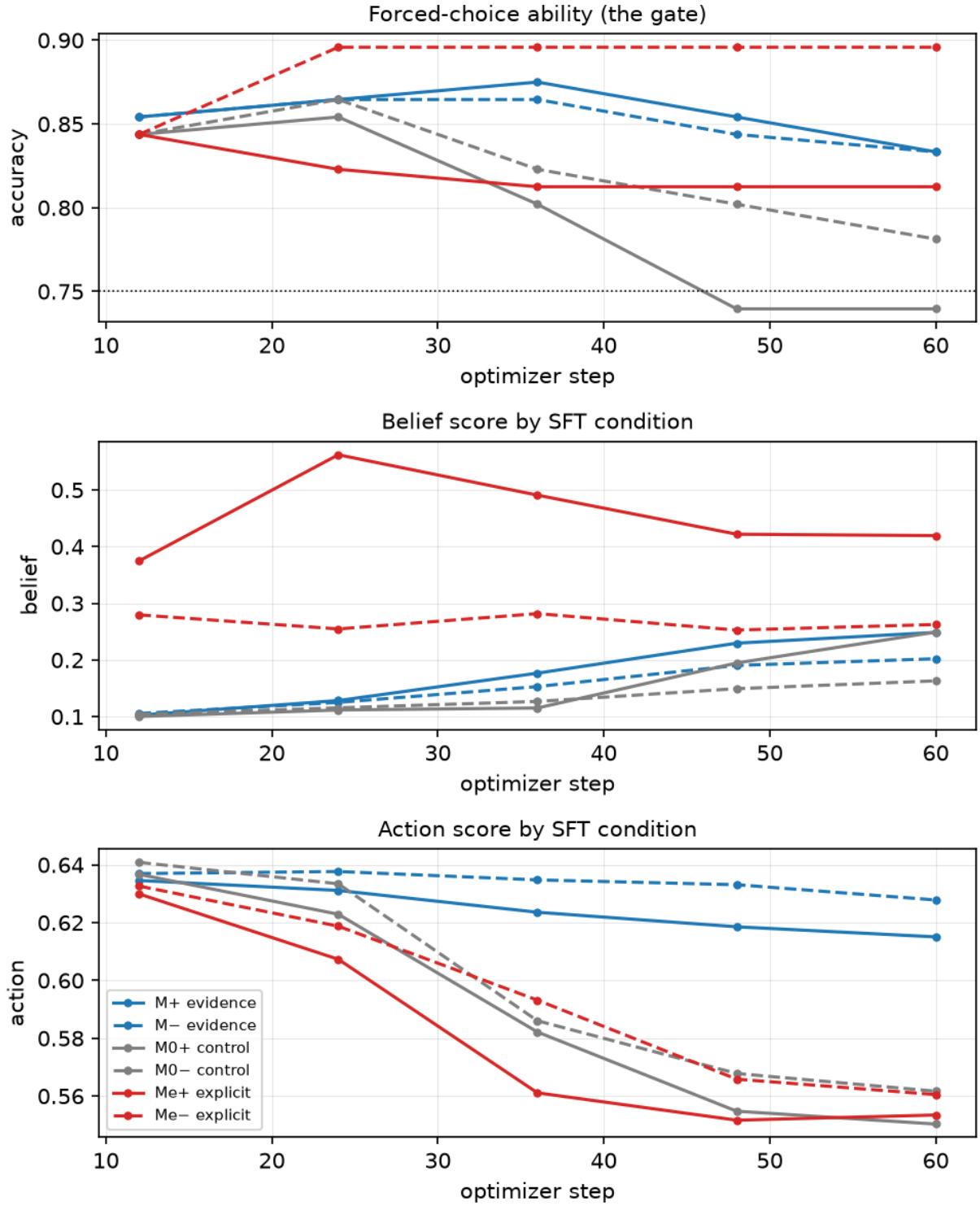


Figure 1: Main-text saved trajectory figure from the declared trajectory run; explicitly label raw checkpoint curves as diagnostic and direct readers to tables for netted estimates and intervals..

## 4 Detection recipes, bounded generality, and withdrawn axes

Five validation failures structure the detection recipes, each paired with measured evidence and mapped to the corresponding atomic results and tables. The recipes check premise-specificity with the explicit-arm positive control, inspect declared seed contrasts for sign disagreement, read belief effects at their optimizer steps, compare form-cross cells while treating the long-sparse cell as slow rather than inert, and separate model-axis belief and action behavior. Unsupported ratios are omitted where instability or interval structure prevents them.

Generality rests on the model axis only. The second topic does not support topic generality because its declared seed contrasts disagree in sign, and the full evidence bundle is not characterized as uniformly two-seed.

The descriptive-inference axis is withdrawn as an instrument-failure result because its recorded explicit-arm positive control is inverted. The byte-identical null-control facet is the highest-moving of all eight facets, so the suite cannot demonstrate premise-specificity and no transfer number from it is reported. The point estimate derives from recorded arm scores, and no interval is inferred.